M7 focused corrections - synthetic issue-rich review

10 evidence-qualified findings across 13 image occurrences; completed execution.

Authored controls demonstrate concurrent integrity problems, declared separation, compatibility blockers and a changed mask. No customer content.

> **COMPLETED** · 13 occurrences. Completion describes audit execution, not dataset acceptance. No universal image-quality score. Similarity does not prove leakage or identify a correct annotation. Images, labels and source paths are excluded unless explicitly disclosed.

Priority review actions

Review integrity and declared separation first, then compatible changes and investigation candidates. Counts below have different populations and must not be added into a quality score.

1. Resolve source and identity problems

deterministic facts

- 1 image locations could not be decoded.
- 2 occurrences have ambiguous logical keys.
- 1 source/reference diagnostic reasons need resolution.

Next action: Inspect the retained local references, repair the declared sources or keys, then re-audit. Dependent comparisons cannot establish a clean result.

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2. Resolve declared split separation

declared requirements

- 4 left-side occurrences repeat exact bytes across a declared separation.
- 5 left-side occurrences share declared groups (per_key matching).
- Group separation has missing values or unresolved key/namespace populations; inspect its disclosed scope.

Next action: Review the overlapping asset/group membership and revise the split assignment or declaration. This evidence does not prove training exposure or evaluation inflation.

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3. Review annotation integrity

facts and compatible differences

- 5 occurrences have invalid or partially valid annotations; reason counts can overlap.
- Some geometric checks are blocked; source views do not establish aligned annotation agreement.
- 1 exact-raster groups have compatible normalized annotation differences.
- 2 differing annotation groups cannot be geometrically compared.

Next action: Inspect supplied states, taxonomy and coordinate frames. Correct declared violations; adjudicate compatible differences locally. The audit does not identify the correct label.

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4. Resolve comparison and coverage limits

changes and coverage boundaries

- Version alignment is blocked; only population and exact-content overlap are available.
- 1 aligned occurrences have compatible annotation changes.

Next action: Review the aligned changes and blocked populations. Supply compatible declarations where possible, or explicitly accept the bounded scope before making a dataset decision.

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5. Investigate visual and repetition evidence

declared quality rules and review candidates

- 6 explicit visual-quality rule violations (not unique-image counts).
- 3 perceptual raster pairs are available for investigation.
- 9 cross-split occurrences with exact byte repetition are available for investigation.

Next action: Apply explicit quality expectations separately from similarity candidates. Inspect examples locally; similarity alone proves neither leakage nor an incorrect label.

Inventory

inventory Examined

deterministic fact

occurrences	13
directory entries	0
image locations	9
physical media files	9
physical alias locations	0

source populations	1 / source: source_4 1 / role: current 1 / occurrences: 1 2 / source: source_3 2 / role: reference 2 / occurrences: 1 3 / source: source_2 3 / role: test 3 / occurrences: 4 4 / source: source_1
logical media bytes	45050
metadata bytes	1379

2 additional rows or details remain in strict JSON. This table is presentation-bounded.

Decode and representations

decode Review

deterministic fact

asset states	decoded: 8 corrupt or truncated: 1
format extension mismatches	0
encoded formats	PNG: 6 JPEG: 2 unknown: 1
modes	RGB: 8 unknown: 1
decoded pixels	23936
dimensions observed	8
frame policy	single frame only

Visual evidence

visual quality Examined

contextual observation

eligible assets	8
exclusions	decoded: 0 corrupt or truncated: 1
contrast	population: 8 minimum: 0 median: 0.14662 maximum: 0.15553
mean luminance	population: 8 minimum: 0 median: 0.4945 maximum: 0.50615
dark fraction	population: 8 minimum: 0 median: 0 maximum: 1
bright fraction	population: 8 minimum: 0 median: 0 maximum: 0
entropy	population: 8 minimum: -0 median: 7.1437 maximum: 7.3036
sharpness	population: 8 minimum: 0 median: 0.42784 maximum: 0.50915
width	population: 8 minimum: 32 maximum: 80
height	population: 8 minimum: 24 maximum: 64
representation	oriented opaque RGB; preview longest side 512; no arbitrary ICC conversion

visual extremes Bounded

contextual observation

interpretation	Contextual extremes in eligible preview representations, not proof of unusability.

quality expectations Review

declared violation

declared violations	min width: 4 min contrast: 2

Annotation integrity

annotations Blocked

deterministic fact

occurrence states	observed: 7 empty: 4 absent: 1 null: 1
validity	valid: 8 partial: 3 invalid: 2
blocked geometric checks	geometric checks blocked: 2 coordinate comparison blocked: 1
supplied status	unknown: 13
invalidity reasons	box out of bounds or nonpositive: 1 required annotation missing: 2 source unavailable: 1 unknown category: 1 declared dimension mismatch: 1
normalized occurrences	13
object visits	5
mask pixels	1920
empty masks	0
segmentation raster pixels	0

Dataset composition

composition Examined

contextual observation

categories	1 / source: source_1 1 / category: category_a757ae39d9aacfc519265530 1 / annotation instances: 2 1 / image occurrences: 2 2 / source: source_2 2 / category: category_c2efe7fffbe322c41fbdcf28 2 / annotation instances: 2 2 / image occurrences: 2 3 / source: source_1 3 / category: category_dbf71c6b29893f8495b6b52d
distinct source categories	5
category detail omissions	0
label cardinality	1: 7 0: 6
splits	1 / split: split_3abf8497c9d52bf961bea9f3 1 / occurrences: 7 2 / split: split_62f05bb64a9a0c84aff792a3 2 / occurrences: 4 3 / split: split_f1b6f2d7cf36b10f61ff74a5 3 / occurrences: 1 4 / split: split_7ed4e9e974d667f21d75df1c 4 / occurrences: 1
box area	population: 3 minimum: 144 maximum: 396 median: 396
crowd objects	0

object locations and sizes	center x fraction / population: 3 center x fraction / minimum: 0.1375 center x fraction / maximum: 0.1875 center x fraction / mean: 0.1625 center y fraction / population: 3 center y fraction / minimum: 0.21875 center y fraction / maximum: 0.25 center y fraction / mean: 0.22917 width fraction / population: 3 width fraction / minimum: 0.225
mask occupancy	population: 2 minimum: 0.1 maximum: 0.1 mean: 0.1
mask occupancy population	Each examined mask independently; overlapping instance areas are not a union.
category count policy	Well-formed supplied members; taxonomy violations remain separately accounted.
group populations	1 / source: source_2 1 / groups declared: Yes 1 / complete group occurrences: 3 1 / incomplete group occurrences: 1 1 / distinct complete groups: 3 2 / source: source_3 2 / groups declared: No 2 / complete group occurrences: Unavailable 2 / incomplete group occurrences: Unavailable 2 / distinct complete groups: Unavailable

26 additional rows or details remain in strict JSON. This table is presentation-bounded.

Duplication candidates

exact duplicates bytes Review

contextual observation

groups	3
participating occurrences	9
cross split occurrences	9

exact duplicates native Review

contextual observation

groups	3
participating occurrences	9
cross split occurrences	9

exact duplicates oriented Review

contextual observation

groups	3
participating occurrences	9
cross split occurrences	9

annotation disagreement Review

contextual observation

compatible exact raster groups with differences	1
payload difference groups blocked	2
correct label identified	No

near duplicates Review

investigation candidate

qualifying raster pairs	3
radius	6
representation	DCT64 opaque RGB preview; seven disjoint partitions
approximate groups are equivalence classes	No

Split and version comparisons

split requirements Review

declared violation

left	split_3abf8497c9d52bf961bea9f3

right	split_62f05bb64a9a0c84aff792a3
left occurrences	7
right occurrences	4
complete group population left	4
complete group population right	3
group mode	per_key
group separation applicability	partial
group keys	1 / key: group_key_ce87acdf4edf79811286a9e0 1 / declared left: 7 1 / declared right: 4 1 / valid left: 5 1 / valid right: 3 1 / comparable left: 5 1 / comparable right: 3 1 / overlap left occurrences: 2 1 / applicability: partial 2 / key: group_key_54688d530841e85a72eb748e
exact asset overlap occurrences	4
declared group overlap occurrences	5
complete declared group overlap occurrences	4
composite group overlap occurrences	0
overlap count unit	Left occurrences, counted once even when multiple keys overlap; asset and group violations are separate reasons.
asset separation required	Yes
group separation required	Yes

8 additional rows or details remain in strict JSON. This table is presentation-bounded.

versions Blocked

deterministic fact

population delta	-3
reference exact content overlap occurrences	4
reference	source_1
current	source_2
reference population	7
current population	4
alignment	blocked
reason	logical_keys_absent_or_ambiguous

versions Examined

contextual observation

population delta	0
reference exact content overlap occurrences	1
reference	source_3
current	source_4
reference population	1
current population	1
added	0
removed	0
alignment	explicit_logical_keys
aligned	1

changes	taxonomy changes: 0 byte changes: 0 native raster changes: 0 oriented raster changes: 0 split changes: 1 group changes: 0 annotation state changes: 0 compatible annotation changes: 1

logical identity Review

deterministic fact

occurrences with duplicate source keys	2

Methods and coverage

Population and search coverage Bounded

method and execution boundaries

execution	completed
source integrity	Verified per-file bytes and membership; no atomic filesystem snapshot.
search interpretation	Bounded candidates do not establish global absence, leakage or annotation correctness.
inventory	required occurrences: 13 inventory complete: Yes
visual quality	eligible: 8 examined: 8 ineligible: 1
visual extremes	eligible: 8 maximum examples per metric end: 5
annotations	occurrences examined: 13 segmentation raster work: 0 segmentation raster limit: 100000000 topology and semantic correctness: not assessed
exact duplicates bytes	representation: bytes eligible occurrences: 12 group detail omissions: 0

exact duplicates native	representation: native eligible occurrences: 11 group detail omissions: 0
exact duplicates oriented	representation: oriented eligible occurrences: 11 group detail omissions: 0
annotation disagreement	variant comparisons are bounded: Yes
versions	aligned: 1 detail omissions: 0
near duplicates	indexed rasters: 5 partition entries: 35 saturated buckets: 0 posting visits: 258 posting rows loaded: 105 posting cache limit bytes: 33554432 hamming comparisons: 10 pair populations / 1 / left: split_3abf8497c9d52bf961bea9f3 pair populations / 1 / right: split_62f05bb64a9a0c84aff792a3 pair populations / 1 / query population: 3 119 further details in strict JSON.

Schema 1.6 · image_reliability_v1 · 2026-10-12T04:35:28.735800+00:00. Tables are presentation-bounded; strict JSON retains the complete bounded analytical evidence and coverage. Resource settings are safeguards, not capacity guarantees.